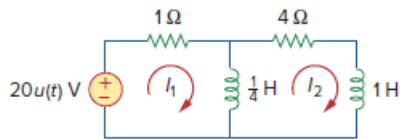


Problem

Solve for the mesh currents in the circuit of Fig. 16.57. You may leave your results in the s -domain.

Figure 16.57:



Step-by-step solution

Step 1 of 6

Refer to figure 16.57 from in the textbook.

Transform the circuit from time domain to s -domain.

Apply the Laplace Transform for the following elements in the circuit

Laplace of unit-step function is,

$$L[u(t)] = \frac{1}{s}$$

Step 2 of 6

For the input voltage element:

$$L[20u(t)] = 20L[u(t)]$$

$$L[20u(t)] = \frac{20}{s}$$

Consider the inductor element of $\frac{1}{4}$ H.

$$sL = s \cdot \frac{1}{4}$$

$$sL = \frac{s}{4}$$

Consider the inductor element of 1 H.

$$sL = s \cdot 1$$

$$sL = s$$

Step 3 of 6

Transform the circuit from time domain to s-domain as shown in figure (a).

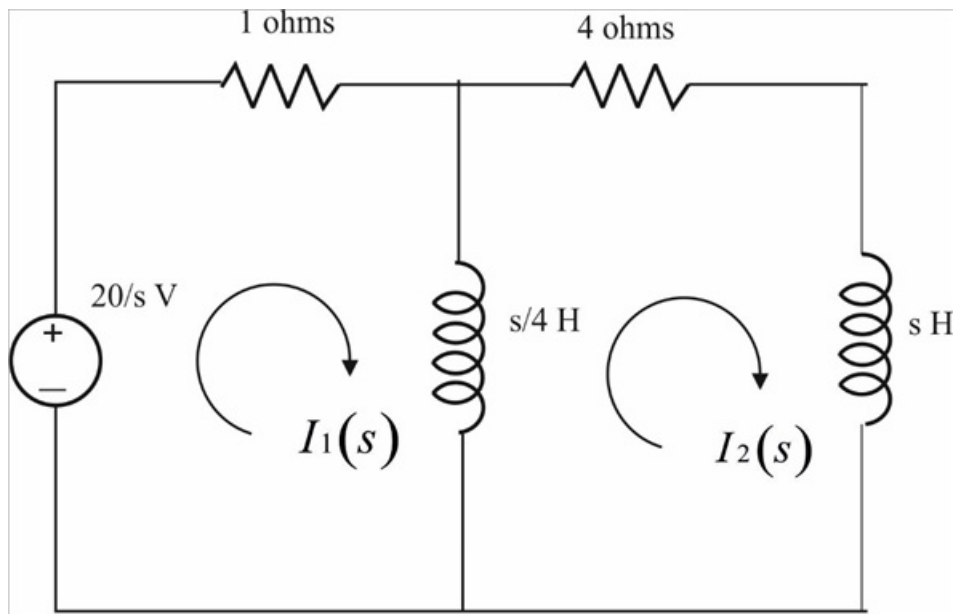


Figure (a)

Step 4 of 6

Apply the Kirchhoff's voltage law to mesh 1 of the circuit shown in figure (a).

$$-\frac{20}{s} + I_1(s) + \frac{s}{4}(I_1(s) - I_2(s)) = 0$$

$$\frac{20}{s} = \left(\frac{s}{4} + 1\right) I_1(s) - \frac{s}{4} I_2(s)$$

$$\left(\frac{s}{4} + 1\right) I_1(s) + \left(\frac{-s}{4}\right) I_2(s) = \frac{20}{s} \dots\dots (1)$$

Apply the Kirchhoff's voltage law to mesh 2 of the circuit shown in figure (a).

$$\frac{s}{4}(I_2(s) - I_1(s)) + 4I_2(s) + sI_2(s) = 0$$

$$\left(\frac{s}{4} + 4 + s\right) I_2(s) = \frac{s}{4} I_1(s)$$

$$I_2(s) = \frac{\frac{s}{4} I_1(s)}{\left(\frac{s + 16 + 4s}{4}\right)}$$

$$I_2(s) = \frac{s}{5s + 16} I_1(s)$$

Step 5 of 6

Recall equation (1).

$$\left(\frac{s}{4}+1\right)I_1(s)+\left(\frac{-s}{4}\right)I_2(s)=\frac{20}{s}$$

Replace $I_2(s)$ by $\frac{s}{5s+16}I_1(s)$.

$$\left(\frac{s}{4}+1\right)I_1(s)+\left(\frac{-s}{4}\right)\frac{s}{5s+16}I_1(s)=\frac{20}{s}$$

$$I_1(s)\left[\frac{s}{4}+1-\frac{s^2}{20s+64}\right]=\frac{20}{s}$$

$$I_1(s)\left[\frac{s(20s+64)+20s+64-4s^2}{4(20s+64)}\right]=\frac{20}{s}$$

$$I_1(s)\left[\frac{20s^2+64s+20s+64-4s^2}{4(20s+64)}\right]=\frac{20}{s}$$

Simplify further,

$$I_1(s)\left[\frac{16s^2+84s+64}{4(20s+64)}\right]=\frac{20}{s}$$

$$I_1(s)\left[\frac{4s^2+21s+16}{(20s+64)}\right]=\frac{20}{s}$$

$$\begin{aligned} I_1(s) &= \frac{20(20s+64)}{s(4s^2+21s+16)} \\ &= \frac{80(5s+16)}{s(4s^2+21s+16)} \end{aligned}$$

Therefore, mesh current $I_1(s)$ is $\boxed{\frac{80(5s+16)}{s(4s^2+21s+16)}}$.

Step 6 of 6

Consider the following relation:

$$\begin{aligned} I_2(s) &= \frac{s}{5s+16}I_1(s) \\ &= \left(\frac{s}{5s+16}\right)\left[\frac{20(20s+64)}{s(4s^2+21s+16)}\right] \\ &= \frac{20(20s+64)}{(5s+16)(4s^2+21s+16)} \\ &= \frac{20 \times 4(5s+16)}{(5s+16)(4s^2+21s+16)} \\ &= \frac{80}{(4s^2+21s+16)} \end{aligned}$$

Therefore, mesh current $I_2(s)$ is $\boxed{\frac{80}{(4s^2+21s+16)}}$.